

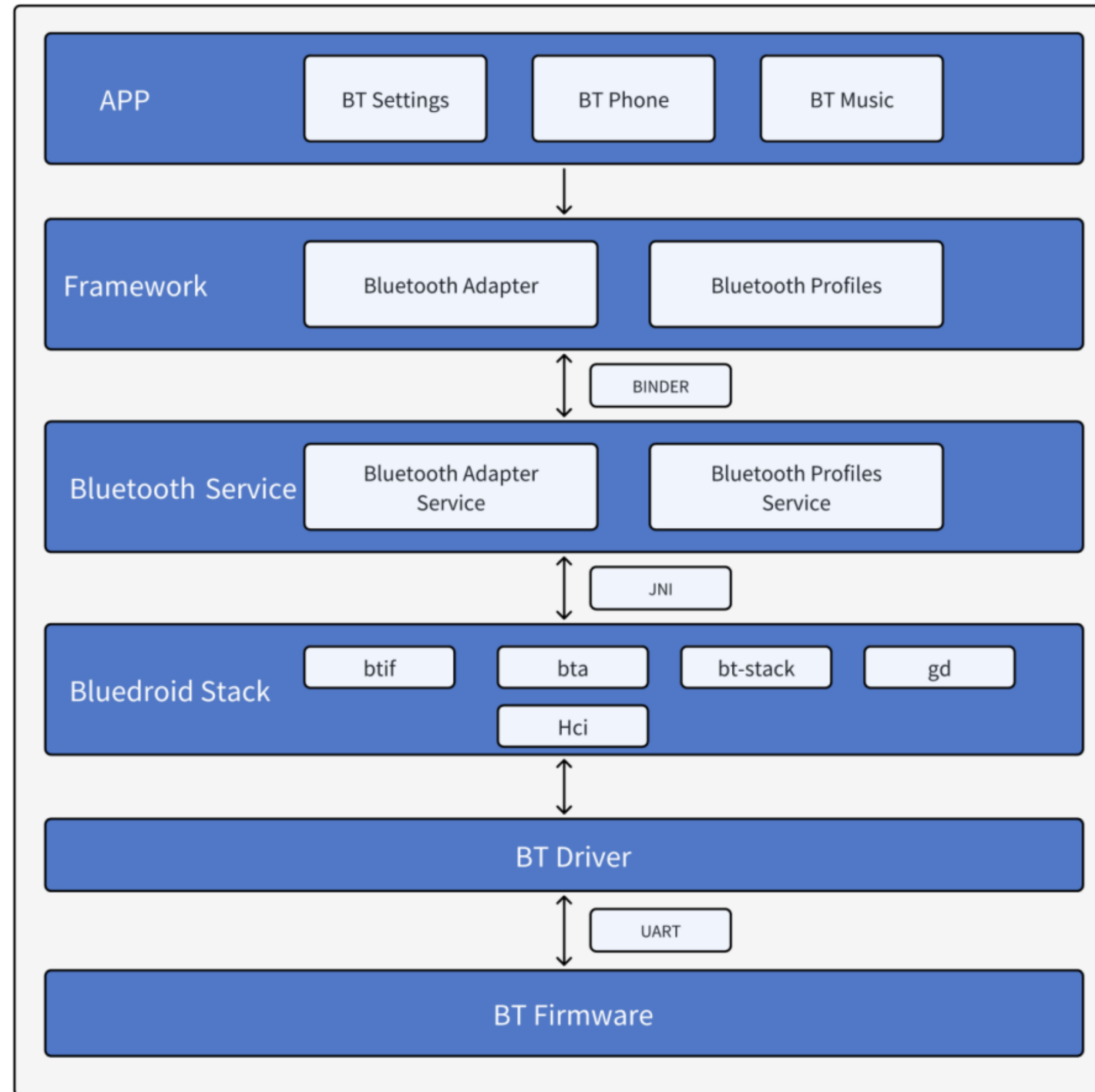


PVT 掌锐电子

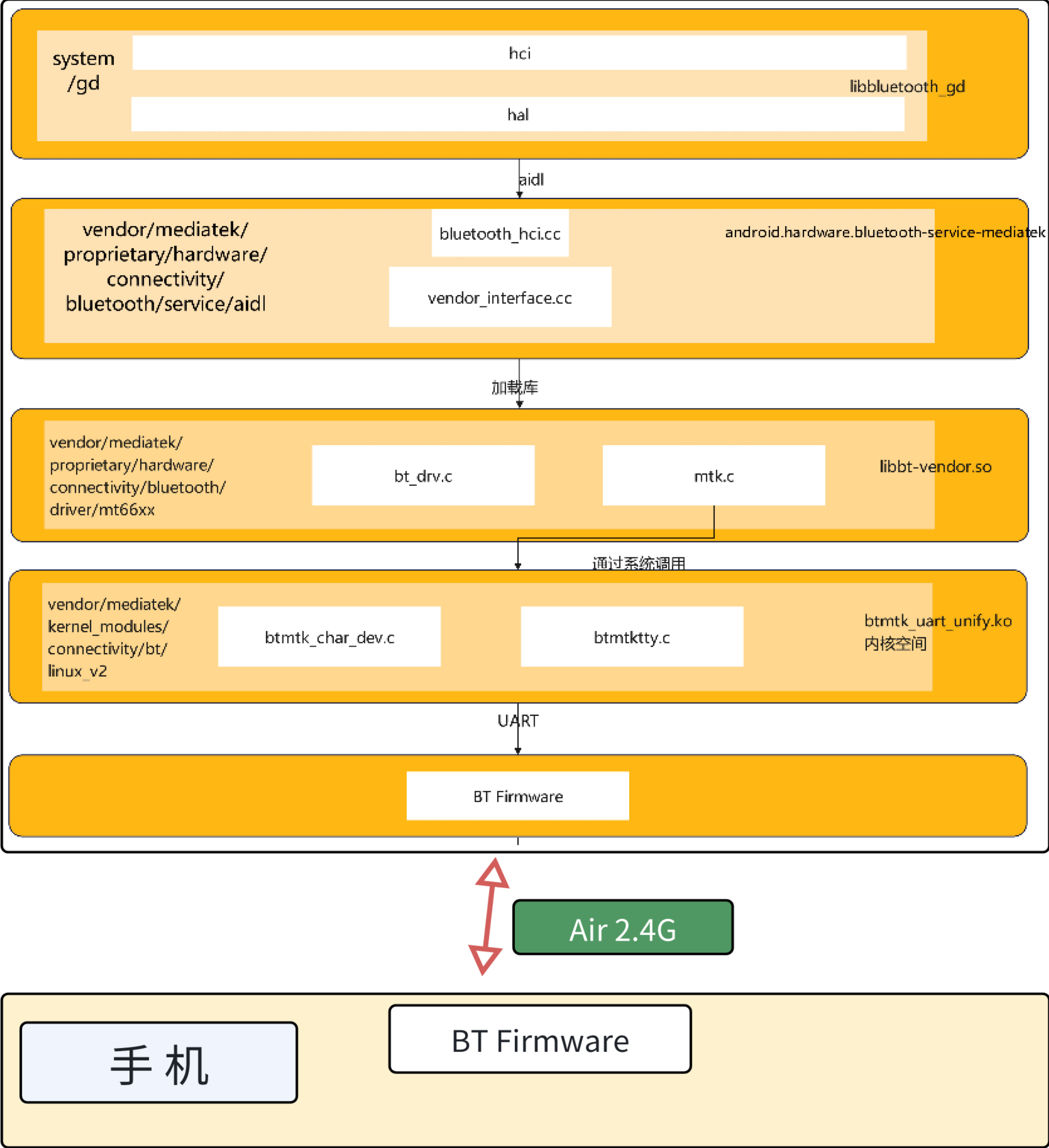


BT安卓架构

蓝牙总体架构



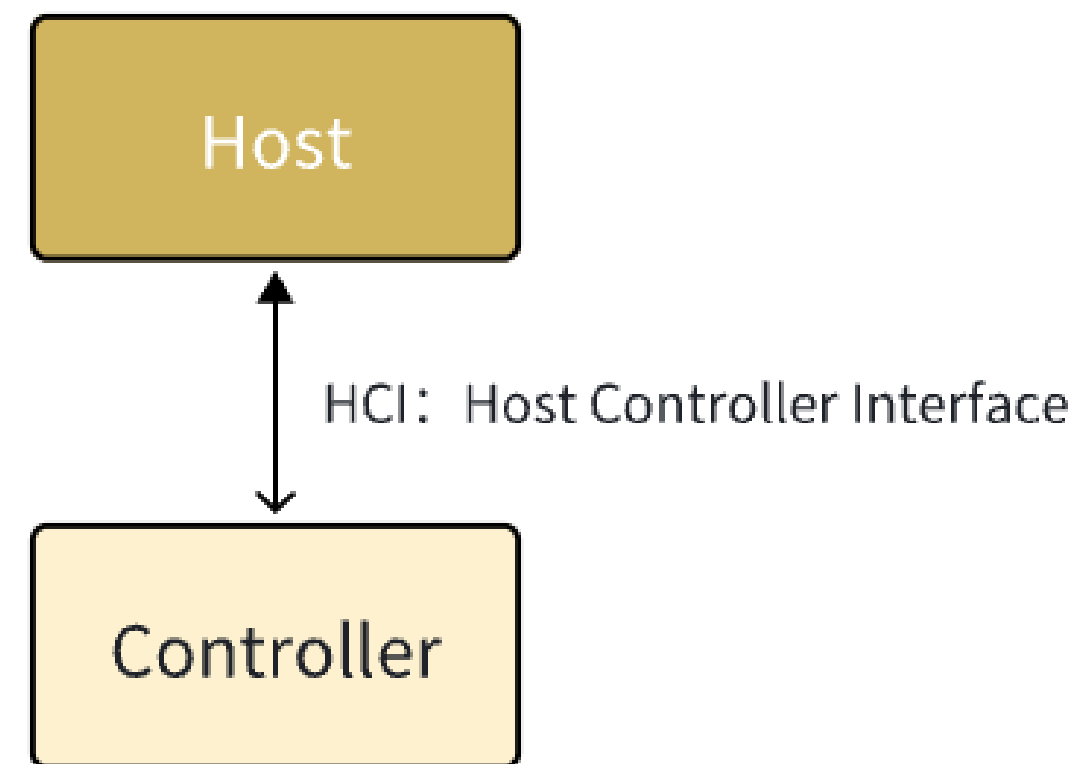
BT HAL架构图



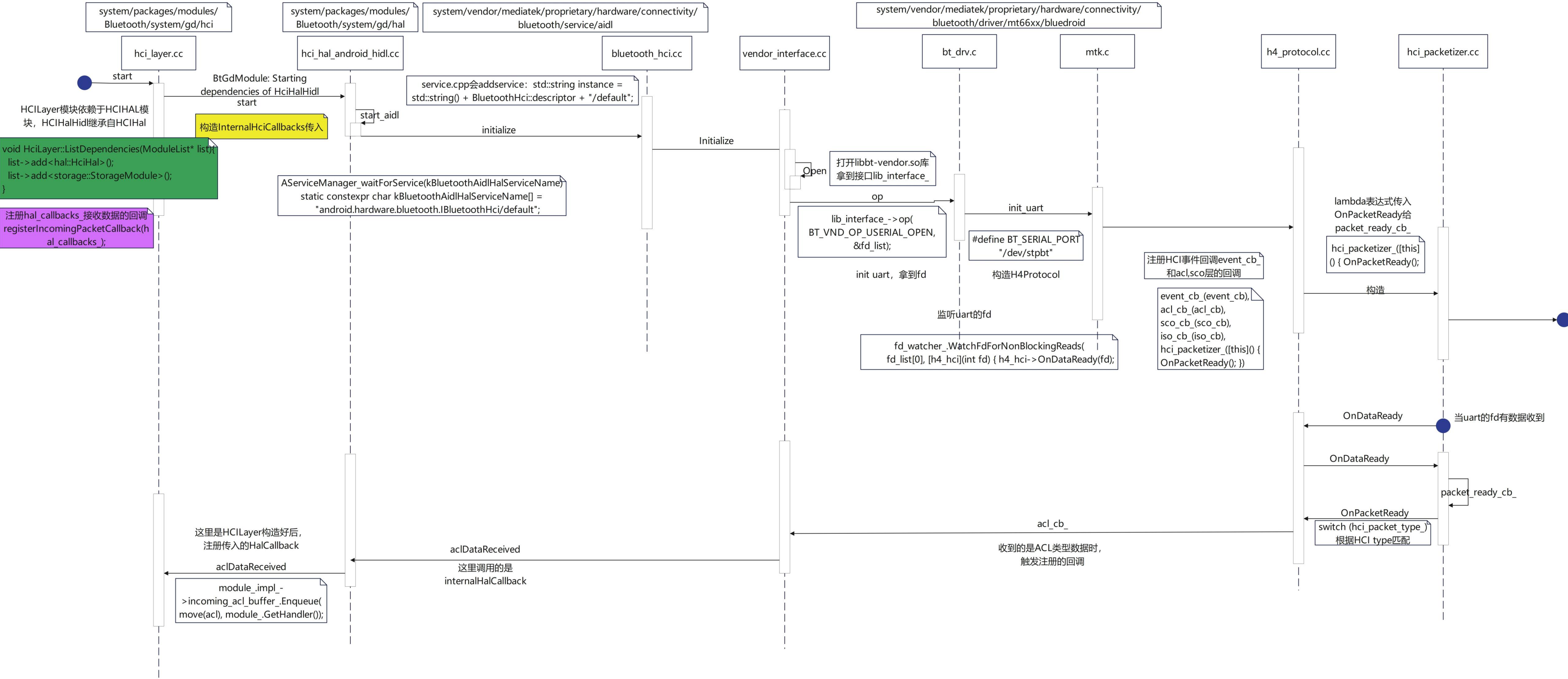
BT Hal

HCI: Host Controller Interface

Host: BT Stack
Controller: Chip Firmware



BT Hal初始化



BT问题定位

BT 问题需要提供的 log:

典型的蓝牙问题有如下一些：连接问题，断线问题，声音问题，NE 问题(蓝牙 Native 进程异常退出)和 Performance 问题，不同问题需要提供的 log 也有所不同，下表是针对不同 case 列出的需要抓的 log。通常情况下，Main log，Kernel log 及 HCI log FW log 都必须提供

M: 必须提供

O: 可以不提供，根据问题的分析情况再提供

C: 有条件的情况下提供或者根据问题的分析情况需要再提供

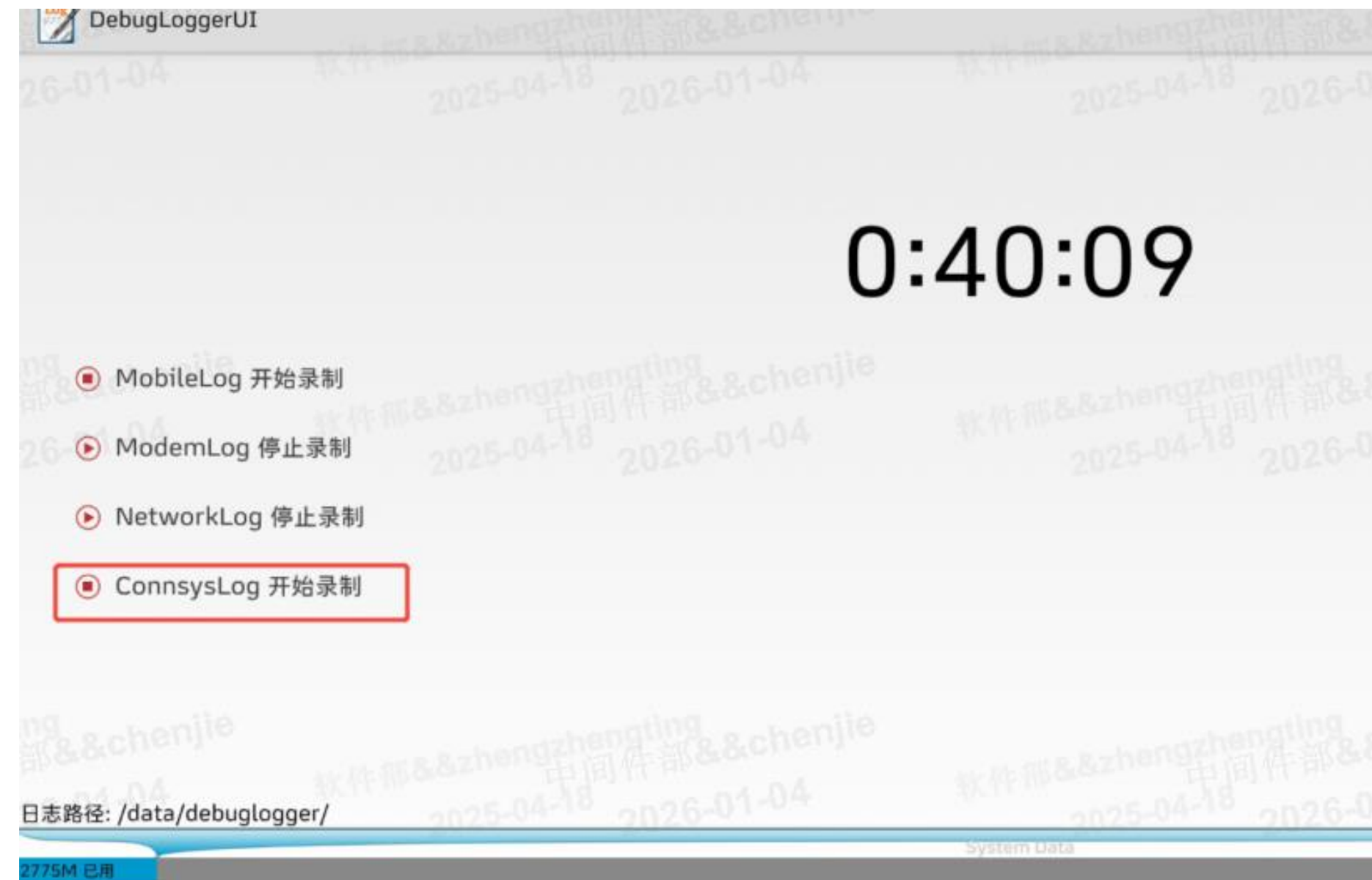
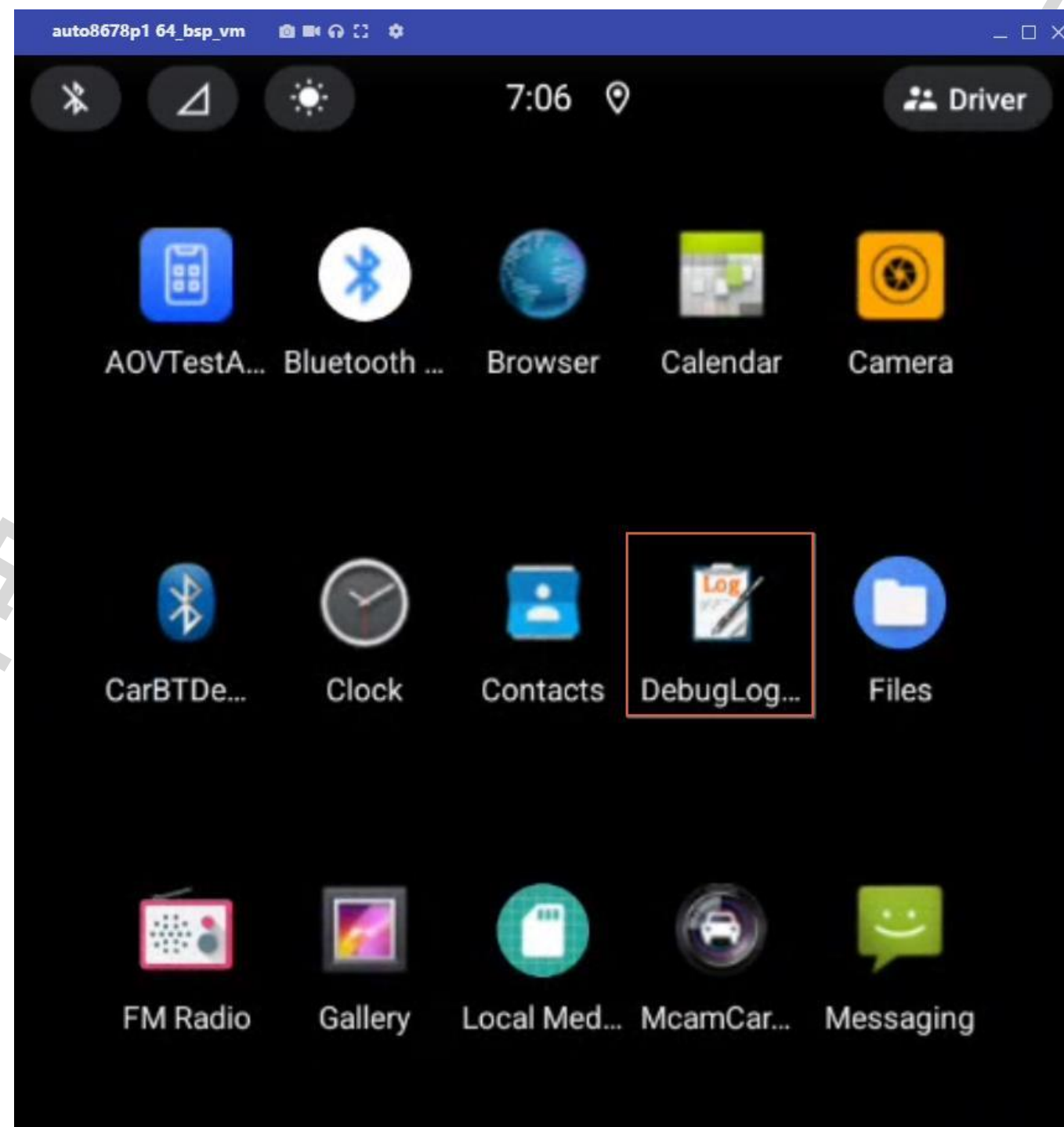
X: 表示通常情况下不需要提供

问题类型	Main log	Kernel log	HCI log	Air log	FW log	Audio dump	DB
连接问题	M	O	M	C	M	X	X
断线问题	M	O	M	C	M	X	X
声音问题	M	O	M	C	M	M	X
NE	M	M	M	C	M	X	M
Performance	M	O	M	M	O	X	X

BT firmware log抓取方法

Debuglogger apk

打开如下开关：



BT HCI log

btsnoop_hci.log						
Files Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help						
Apply a display filter ... <Ctrl-/>						
No.	Time	Source	Destination	Protocol	Leng	Info
→ 1	2020-01-01 00:01:05.829255	host	controller	HCI_CMD	4	Sent Reset
← 2	2020-01-01 00:01:05.848316	controller	host	HCI_EVT	7	Rcvd Command Complete (Reset)
3	2020-01-01 00:01:05.853463	host	controller	HCI_CMD	12	Sent Set Event Mask
4	2020-01-01 00:01:05.854749	controller	host	HCI_EVT	7	Rcvd Command Complete (Set Event Mask)
5	2020-01-01 00:01:05.854945	host	controller	HCI_CMD	12	Sent Set Event Mask Page 2
6	2020-01-01 00:01:05.856155	controller	host	HCI_EVT	7	Rcvd Command Complete (Set Event Mask Page 2)
7	2020-01-01 00:01:05.856314	host	controller	HCI_CMD	6	Sent Write LE Host Supported
8	2020-01-01 00:01:05.857233	controller	host	HCI_EVT	7	Rcvd Command Complete (Write LE Host Supported)
9	2020-01-01 00:01:05.857405	host	controller	HCI_CMD	4	Sent Read Local Name
10	2020-01-01 00:01:05.858515	controller	host	HCI_EVT	255	Rcvd Command Complete (Read Local Name)
11	2020-01-01 00:01:05.858738	host	controller	HCI_CMD	4	Sent Read Local Version Information
12	2020-01-01 00:01:05.860027	controller	host	HCI_EVT	15	Rcvd Command Complete (Read Local Version Information)
13	2020-01-01 00:01:05.860182	host	controller	HCI_CMD	4	Sent Read Local Supported Commands
14	2020-01-01 00:01:05.861290	controller	host	HCI_EVT	71	Rcvd Command Complete (Read Local Supported Commands)
15	2020-01-01 00:01:05.861445	host	controller	HCI_CMD	4	Sent LE Read Local Supported Features
16	2020-01-01 00:01:05.863125	controller	host	HCI_EVT	15	Rcvd Command Complete (LE Read Local Supported Features)
17	2020-01-01 00:01:05.864127	host	controller	HCI_CMD	4	Sent LE Read Supported States
18	2020-01-01 00:01:05.865408	controller	host	HCI_EVT	15	Rcvd Command Complete (LE Read Supported States)
19	2020-01-01 00:01:05.865569	host	controller	HCI_CMD	5	Sent Read Local Extended Features
20	2020-01-01 00:01:05.876882	controller	host	HCI_EVT	17	Rcvd Command Complete (Read Local Extended Features)
21	2020-01-01 00:01:05.878069	host	controller	HCI_CMD	5	Sent Read Local Extended Features
22	2020-01-01 00:01:05.882165	controller	host	HCI_EVT	17	Rcvd Command Complete (Read Local Extended Features)
23	2020-01-01 00:01:05.882386	host	controller	HCI_CMD	5	Sent Read Local Extended Features
24	2020-01-01 00:01:05.883829	controller	host	HCI_EVT	17	Rcvd Command Complete (Read Local Extended Features)
25	2020-01-01 00:01:05.884080	host	controller	HCI_CMD	12	Sent LE Set Event Mask
26	2020-01-01 00:01:05.894857	controller	host	HCI_EVT	7	Rcvd Command Complete (LE Set Event Mask)
27	2020-01-01 00:01:05.895070	host	controller	HCI_CMD	4	Sent Read Buffer Size

BT Firmware log使用场景和抓取方法

1、使用场景

该log主要用于定位蓝牙底层固件问题，关键字如fw_assert，常见场景如下：

- (1) 连接问题：当设备连接断线或者连接异常超时场景下，分析异常或者耗时情况时，指向固件执行报错
- (2) NE问题：当NE或KE报错指向固件报错时
- (3) 蓝牙电话声音异常：通过audio dump，音频传输链路上指向固件传输过来的数据本身有异常，也能通过FW log体现出来

2、操作方法（包括说明需要提交给研发什么结果）

1、打开开关，复现问题

2、在机器的如下路径确认是否有抓到log：
会以BT_FW_时间戳.clog的形式抓出来



- 1. 打开 mtklogger 工具
- 2. 点击 ConnsysLog
- 3. 点击右上角的菜单
- 4. 进入 ConnsysLog Settings 界面
- 5. 将 BT Firmware Log Level 改为 Debug。

```
CUX_V5L_8676_VT252:/data/debuglogger/connsyslog/fw/CsLog_2025_0630_170801 # ls
BTMCU_FW_2025_0630_170801.clog.curf BT_FW_2025_0630_170801.clog.curf ICS_FW_20
CUX_V5L_8676_VT252:/data/debuglogger/connsyslog/fw/CsLog_2025_0630_170801 #
```

fw log抓取方法示例

三方协议栈抓取方法示例：

1 hci log通过如下adb命令打开：

```
adb root
adb remount
adb shell
setprop persist.bluetooth.btsnoopenable true
```

2 fw log打开方法：

先将车机里/system/etc/bluetooth/nForeBluetooth.properties这个文件通过adb pull出来，把MtkEnable=false这行改为MtkEnable=true

再将修改后的nForeBluetooth.properties文件push进/system/etc/bluetooth/目录替换原来的文件

```
adb root
adb remount
adb push (你的存放文件的电脑路径)/nForeBluetooth.properties /system/etc/bluetooth/
```

3 上述步骤全部配置完成后，断电重启，进系统设置关闭再打开一下蓝牙，然后开始复现问题。

复现问题后抓取log：fw log是直接打印在hci log里面的，所以复现后，记录出问题的时间点，并把/data/misc/bluetooth/logs目录adb pull出来，复现到问题后5分钟内抓出，以免太久log被冲掉

PVT

Thank You

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